DO NOW — RECALL

Lesson 100 · GPUs: Parallel Work Beyond Graphics

Name: _____ Class: _____ Date: _____

1. Which processor is designed for throughput on massive parallel workloads?

2. What is parallelism?

3. What is a data dependency?

4. An image array has 4 rows × 3 columns. How many pixels?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

GPU.

2. Model answer

Actually running independent work at same time.

3. Model answer

A task needs another task's result.

4. Model answer

12.

By the end, you can…

• GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

• Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

• Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Vocabulary — know these terms

GPU — Graphics Processing Unit.

SIMT — Single instruction, multiple threads execution model.

throughput — Total work per time.

divergence — Threads follow differing control paths.

NEW LEARNING

Read, annotate and explain

Observe

GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

Apply

Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

Evaluate

Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Explain why processing one million independent pixels may suit a GPU.

Follow the reasoning

Each pixel can receive similar operation in parallel; many GPU threads process different positions.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain two features that make a GPU useful for applying same filter to millions of pixels. [6]

Question 2

Give two non-graphics GPU applications and explain how parallelism helps. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare CPU and GPU for 12-item sequential task versus million-element independent matrix operations. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A GPU is used exclusively for drawing computer-game graphics.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.